SOFTWARE ENGINEERING

CSA1012

LAB EXERCISES

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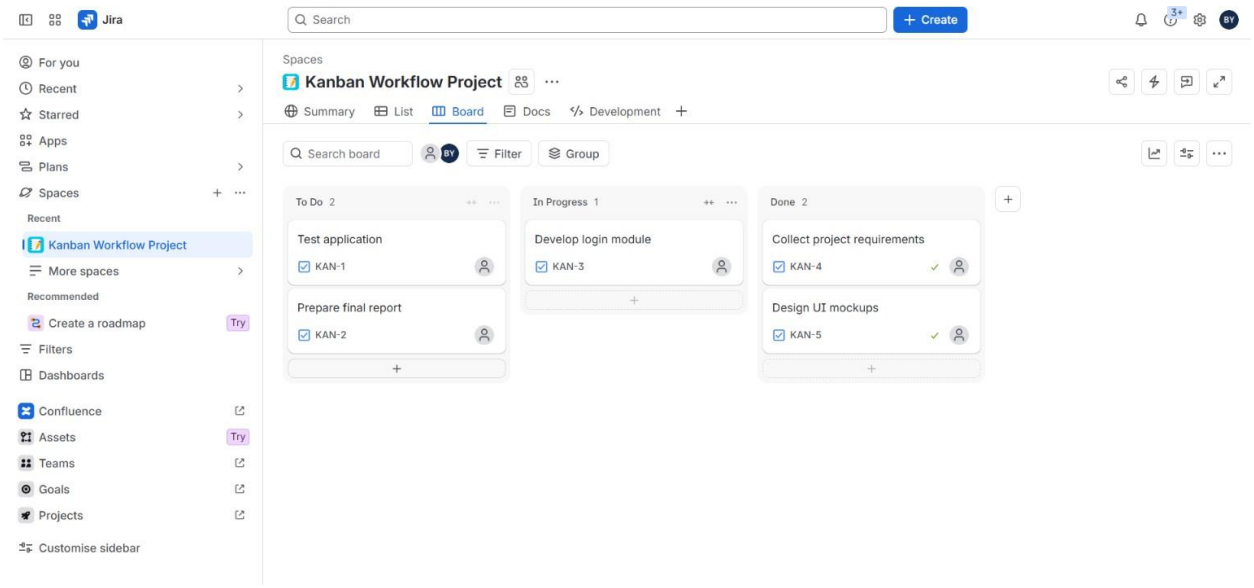
Reg no :1925260163

1)

Sl. No	Question
1	Create a Kanban Board to Visualize the Tasks. • Create Columns for To Do, In-Progress and Done. • Add Atleast 5 Sample Tasks • Move the Tasks across the Columns to Simulate the Workflow.

Tools Used : Jira or Trello

Output :

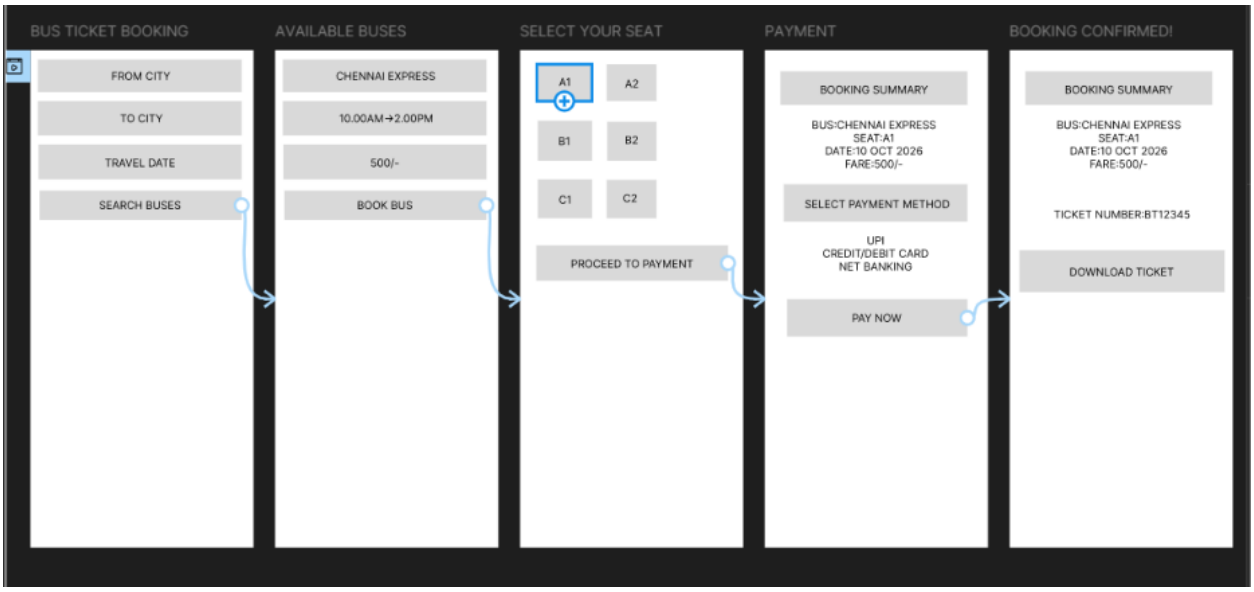


2)

2	Sketch a Simple Prototype of a Bus Ticket Booking System using Figma Tool

Tools Used : Figma

Output :

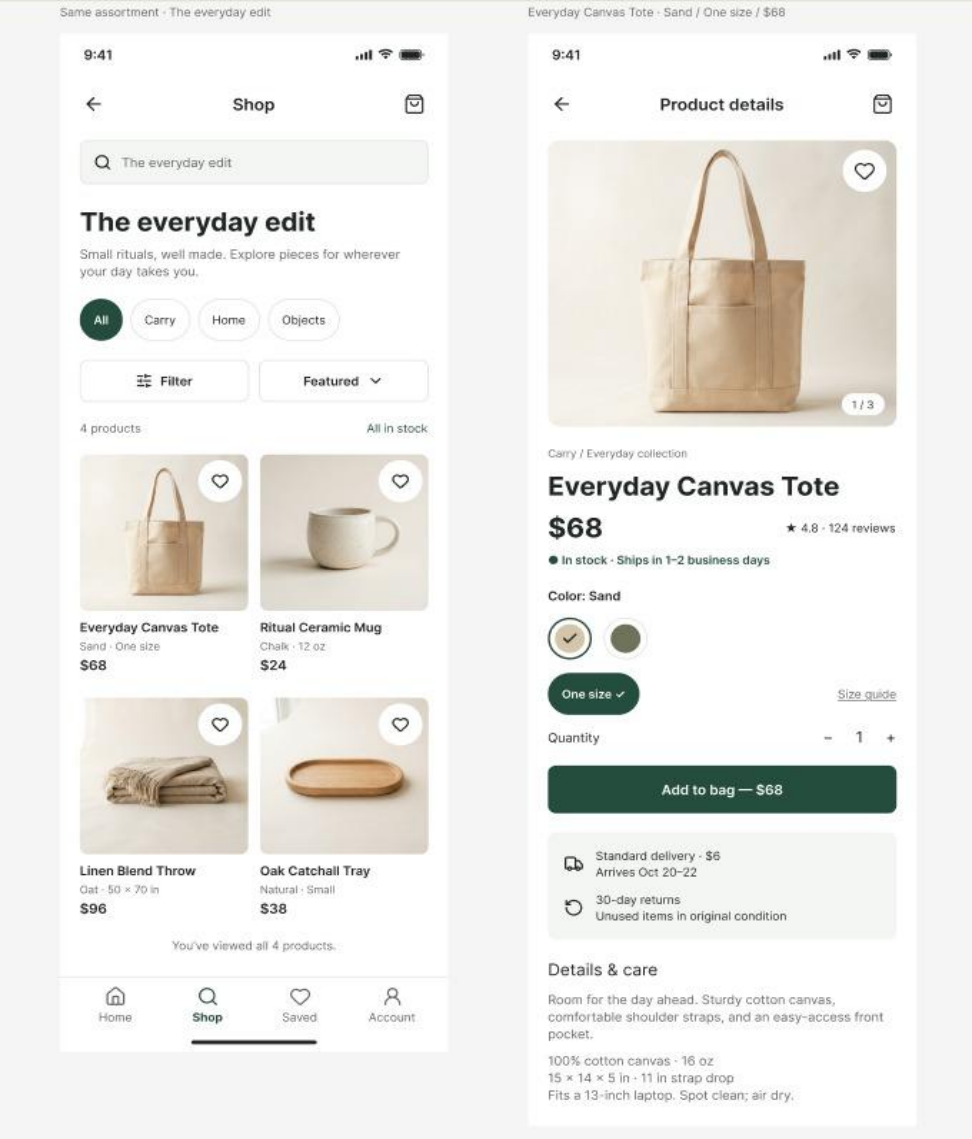


3)

3	The stakeholders have conflicting views on the user interface design for an E-Commerce mobile app. Create a prototype using Figma tool to discuss with the stakeholders to get their feedback and approval.

Tools Used : Figma

Output :

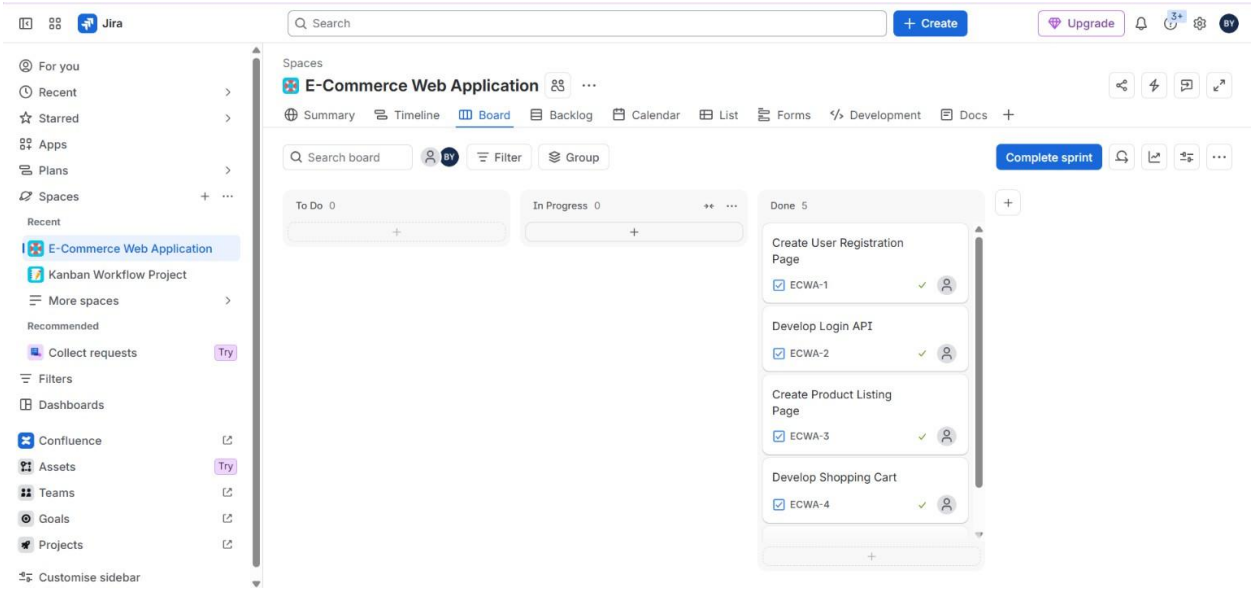


4)

4	Create a Scrum Project in Jira. • Add a backlog with at least 5 items (e.g., "Create user registration page", "Develop API for login"). • Prioritize the backlog and create a 1-week sprint. • Move backlog items into the sprint and start the sprint. • Finally show the Screenshot of the sprint board at the start and end of the sprint.

Tools Used : Jira

Output :



5)

5	Use the following requirements for a Library Management System: • Add a feature to search books by title and author. • Implement an online book reservation system. • Generate monthly reports on borrowed books for administrators. • Enable email notifications for overdue books. • Add support for QR code scanning for borrowing and returning books. • Create a user-friendly dashboard for librarians. • Allow users to review and rate books. • Integrate a chatbot for user assistance. • Develop a mobile app version of the system. • Provide multi-language support. Categorize each requirement using MOSCOW Method (Must-Have, Should-Have, Could-Have, or Won't-Have) based on the following criteria: • Impact on the users and stakeholders. • Feasibility considering time, budget, and resource constraints. Finally Submit the completed Google Sheet or Excel file with all requirements categorized and justified.

Tools Used : Google sheet or Excel

Output:

No.	Requirement	MoSCoW Category	Justification: User / Stakeholder Impact	Justification: Feasibility (Time, Budget, Resources)
1	Search books by title and author	**Must-Have**	Core library function that helps users find books quickly and supports everyday borrowing.	Highly feasible; a basic search can be implemented using existing catalogue fields.
2	Online book reservation system	**Must-Have**	Allows users to reserve unavailable or desired books and improves access to library resources.	Feasible with a reservation queue and availability checks; essential workflow can be built within a typical project.
3	Monthly reports on borrowed books for administrators	**Should-Have**	Helps administrators monitor borrowing trends and prepare monthly summaries, but the system can operate without automated reports initially.	Moderately feasible; requires report queries and export/formatting, which can follow core circulation features.
4	Email notifications for overdue books	**Should-Have**	Reminds users to return books and helps reduce overdue borrowing, though staff can initially follow up manually.	Feasible if an email service is available; requires scheduled checks and email configuration.
5	QR-code scanning for borrowing and returning books	**Could-Have**	Can speed up circulation and reduce manual entry, but standard staff-assisted checkout can work without it.	Additional device/camera integration and testing are needed; can be added after the core system.
6	User-friendly dashboard for librarians	**Must-Have**	Gives librarians a clear way to manage books, loans, reservations, and users; strongly affects staff productivity.	Feasible as a basic dashboard using standard interface components; advanced customization can be deferred.

GITHUB LINK :

https://github.com/moulikasai592-ux/CSA1012-SOFTWARE-ENGINEERING-